Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	*(a)*	(i)	Acceptable line of best fit by eye – bisects data points at (0.4,0.925) and (0.5,1.25) (1) Doesn't need to pass through zero but needs to be drawn back to an axis Gradient calculated. Accept range: 2.3 – 2.5 (1) No unit penalty		2		2	2	2
		(ii)	Unit of force (Newton) seen as kg m s^{-2} (1) $\frac{\text{m s}^{-2}}{\text{kg m s}^{-2}}$ seen [= kg^{-1}] (1) Accept: Gradient = $\frac{a}{F} = \left(\frac{\text{N}}{\text{kg N}}\right)$ argument award 2 marks	1	1		2	1	2
		(iii)	$\frac{1}{\text{gradient}}$ = 0.40 – 0.44 [kg] (1) (**ecf** on gradient). Accept correct use of data points. 0.42 = M + 0.06 (1) M = 0.36 [kg] (1) Accept range: 0.34 – 0.38 [kg]			3	3	3	3
		(iv)	Quality: Points lie close to line of best fit so quality acceptable or reference to possible anomaly at 0.5 N (1). Don't accept reference to measuring apparatus. Sufficiency: Too few points plotted **or** more needed between 0.0 and 0.4 N or results missing at e.g. 0.1 N or 0.3 N (1)			2	2		2
	(b)	(i)	Weight of glider = 9.81 × M (= 3.1 – 3.7) N (**ecf on M**) (1) Force due to air must be **same answer** as weight of glider (1)		2		2	1	
		(ii)	Forces act on the same body (1) Forces are not of the same kind **or** details provided e.g. contact/electrostatic and gravitational (1)	2			2		
			Question 3 total	**3**	**5**	**5**	**13**	**7**	**9**